CS & IT ENGINEERING

Operating System

Process Synchronization

Lecture -3

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Recap of Previous Lecture

doubt

Topic Critical Section

Topic Mutual Exclusion

Topics to be Covered

Topic Peterson’s Solution

Topic Hardware Solutions of Synchronization

Topic Test-And-Set()

Topic Swap()

M.E. X
Progress ✓
B.W. X

Solution 1

```
Boolean lock=false;

while(true)
{
    while(lock);
    lock=true;
        //CS
    lock=false;
    RS;
}
```

```
while(true)
{
    while(lock);
    lock=true;
        //CS
    lock=false;
    RS;
}
```

Strict alternation

Solution 2

shared b/w P_0 and P_1

```
int turn=0;
```

✓ Mutual Exclusion

✗ Progress

✓ Bounded waiting

P_0

```
while(true)
{
    while(turn!=0);
        CS
    turn=1;
    RS;
}
```

turn = $\cancel{0}$ 1

P_1

```
while(true)
{
    while(turn!=1);
        CS
    turn=0;
    RS;
}
```

P_0 can enter into C.S. when turn 0.

P_1 can enter into C.S. when turn is 1.

Peterson's Solution

To announce if a process wants to enter into C.S.

```
Boolean Flag[2]={False, False};
int turn;
```

Priority

✓ M.E.

✓ Progress

✓ B.W.

P_0

```
while(true) {
  Flag[0]=true;
  turn=1;
  while(Flag[1] && turn==1);
    CS
  Flag[0]=False;
    RS;
}
```

Flag[0] = ~~F~~ ~~T~~ ~~F~~ T

Flag[1] = ~~F~~ T

turn = ~~1~~ ~~0~~ 1

P_1

```
while(true){
  Flag[1]=true;
  turn=0;
  while(Flag[0] && turn==0);
    CS
  Flag[1]=False;
    RS;
}
```

H.W.

Question 1

```
turn=0;

while(true)
{
    while(turn);
    turn=1;
        //CS
    turn=0;
    RS;
}
```

```
while(true)
{
    while(turn);
turn lock=1;
        //CS
turn lock=0;
    RS;
}
```

H.W.

Question 2

```
lock=False;

while(true)
{
    while(lock!=False);
    CS
    lock=True;
    RS;
}
```

```
while(true)
{
    while(lock!=True);
    CS
    lock=False;
    RS;
}
```

Topic : Synchronization Hardware

→ CPU supports some instructions to provide synchronization.

1. TestAndSet()
2. Swap()

Topic : TestAndSet()

Returns the current value flag and sets it to true.

TestAndSet(flag);

if flag = f

⇒ returns false and flag = T

if flag = T

returns True and Flag = T

Topic : TestAndSet()

✓ M.E.
✓ Progress
✗ B.W.

```
Boolean Lock=False;   True True

boolean TestAndSet(Boolean *trg){
boolean rv = *trg;
*trg = True;
Return  rv;
}
```

P0   P1

```
while(true)
{
 while(TestAndSet(&Lock));
       CS
 Lock=False;
}
```

Topic : Swap()

for each process key variable is personal

```
Boolean Key;                    //Local
Boolean Lock=False;             //Shared

void Swap(Boolean *a, Boolean *b)
{
boolean temp = *a;
*a=*b;
*b=temp;
}
```

P_0 | P_1
key = ~~T~~ T | key = ~~T~~ f

lock = ~~f~~ ~~T~~ T

```
while(true){
  Key = True;
  while (key==True)
     { Swap(&Lock, &Key); }
     CS
  Lock=False;
}
```

✓ M. E.

✓ Progress

✗ B. W.

2 mins Summary

Topic Peterson's Solution

Topic Hardware Solutions of Synchronization

Topic Test-And-Set()

Topic Swap()

Happy Learning

THANK - YOU